1 **(a)** 3 **[B1]**

(b) 10^{-2} or $\dfrac{1}{100}$ **[M1]**

0.01 **[A1]**

(c) $\log_5 125$ **[B1]**

[Total Marks: 4]

2 **(a)** $2x = \dfrac{\log 50}{\log 3}$ **[M1]**

$2x = 3.56\ldots$ **[A1]**

1.78 **[A1]**

(b) $\log y = x - 4$ **[M1]**

$\log y + 4$ **[A1]**

[Total Marks: 5]

3 **(a)** **(i)** 250 **[B1]**

(ii) $250 \times 0.92^t = 50$ or $0.92^t = 0.2$ **[M1]**

$t = \dfrac{\log 0.2}{\log 0.92}$ **[M1]**

19.3 **[A1]**

(b) $800 \times 1.06^n > 2000$ or $1.06^n = 2.5$ **[M1]**

$n = \dfrac{\log 2.5}{\log 1.06}$ $(= 15.7\ldots)$ **[M1]**

16 **[A1]**

[Total Marks: 7]